12.70

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Problem Statement

Question : X bullocks and Y tractors take 8 days to plough a field. If we halve the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

Number of days and Work

The total work done = W

Work done by 1 bullock in one day = x

Work done by 1 tractor in one day = y

Given,

$$\begin{pmatrix} 8X & 8Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = W \tag{3.1}$$

$$\begin{pmatrix} \frac{5X}{2} & 10Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = W \tag{3.2}$$

The combined matrix is

$$\begin{pmatrix} 8X & 8Y \\ \frac{5X}{2} & 10Y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} W \\ W \end{pmatrix} \tag{3.3}$$

Forming the augmented matrix ,

$$\left(\begin{array}{cc|c} 8X & 8Y & W \\ \frac{5X}{2} & 10Y & W \end{array}\right) \tag{3.4}$$

Using Gaussian Eliminations

$$\left(\begin{array}{cc|c} 8X & 8Y & W \\ \frac{5X}{2} & 10Y & W \end{array}\right) \xleftrightarrow{R_2 \to 4R_2 - 5R_1} \left(\begin{array}{cc|c} 8X & 8Y & W \\ -30X & 0 & -W \end{array}\right) \tag{3.5}$$

$$30Xx = W \tag{3.6}$$

Number of days taken $= 30$

C Code

```c
#include <stdio.h>

double days_for_bullocks_alone(double X, double Y) {
    // From equations:
    // 8*(X*b + Y*t) = 1
    // 5*((X/2)*b + 2*Y*t) = 1
    // Solve for b, t symbolically and return 1/(X*b)
    double a1 = 8 * X;
    double b1 = 8 * Y;
    double c1 = 1;

    double a2 = 5 * (X / 2.0);
    double b2 = 5 * (2.0 * Y);
    double c2 = 1;

    // Expressing equations in rate form:
    // (a1/a1)*b + (b1/a1)*t = c1/a1
    // (a2/a1)*b + (b2/a1)*t = c2/a1
```

```
    double det = a1 * b2 - a2 * b1;
    double b_rate = (c1 * b2 - c2 * b1) / det;
    double t_rate = (a1 * c2 - a2 * c1) / det;

    // time for X bullocks alone = 1 / (X * b_rate)
    double T = 1.0 / (X * b_rate);
    return T;
}
int main() {
    double X, Y;
    printf("Enter number of bullocks (X): ");
    scanf("%lf", &X);
    printf("Enter number of tractors (Y): ");
    scanf("%lf", &Y);

    double T = days_for_bullocks_alone(X, Y);
    printf("Days taken by %g bullocks alone = %.2f days\n", X, T)
        ;

    return 0;
}
```

call.py

```python
import ctypes

# Load the shared object file (make sure plough.so is in the same
    directory)
plough = ctypes.CDLL('./plough.so')

# Define argument and return types for the C function
plough.days_for_bullocks_alone.argtypes = [ctypes.c_double,
    ctypes.c_double]
plough.days_for_bullocks_alone.restype = ctypes.c_double

def main():
    print(" Bullocks and Tractors Ploughing Problem ")
    # Get input values
    X = float(input("Enter number of bullocks (X): "))
    Y = float(input("Enter number of tractors (Y): "))

    # Call the C function
    T = plough.days_for_bullocks_alone(X, Y)
```

```python
    # Display result
    print(f"\n Days taken by {X:.0f} bullocks alone to plough the field = {T:.2f} days")

if __name__ == "__main__":
    main()
```